

Type I.

1. If $A + B + C = 180^\circ$, prove that

(i) $\cos^2 A + \cos^2 B + \cos^2 C + 2\cos A \cos B \cos C = 1$

(ii) $\sin^2 A + \sin^2 B + \sin^2 C = 2(1 + \cos A \cos B \cos C)$ [P. U. 45, 48]

(iii) $\cos^2 A + \cos^2 B - \cos^2 C = 1 - 2\sin A \sin B \cos C$

(iv) $\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2} - \cos^2 \frac{C}{2} = 2\cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$

(v) $\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2} + \cos^2 \frac{C}{2} = 2 + 2\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

2. If $A + B + C = \frac{\pi}{2}$, show that

(i) $\sin^2 A + \sin^2 B + \sin^2 C = 1 - 2\sin A \sin B \sin C$

(ii) $\cos^2 A + \cos^2 B + \cos^2 C = 2 + 2\sin A \sin B \sin C$

3. If $\alpha + \beta + \gamma = 2\pi$, prove that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma - 2\cos\alpha \cos\beta \cos\gamma = 1$$

4. (i) If $\alpha + \beta = \gamma$, prove that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + 2\cos\alpha \cos\beta \cos\gamma \quad [\text{B. U. 55; P. U. 56}]$$

(ii) If $\alpha + \beta = 60^\circ$, prove that $\cos^2\alpha + \cos^2\beta - \cos\alpha \cos\beta = \frac{3}{4}$

[S. S. 58, 69, 72]

(iii) Show that $\cos^2\theta + \cos^2(\alpha + \theta) - 2\cos\alpha \cos\theta \cos(\alpha + \theta)$ is independent of θ .

[Hint : Put $\beta = \theta$ and $\gamma = \alpha + \theta$ in 4 (i)]

(iv) If $A + B + C = \pi$ and $A + B = 2C$, prove that

$$4(\sin^2A + \sin^2B - \sin A \sin B) = 3$$

[P. U. 55]

[Hint : $A + B + C = \pi$ and $A + B = 2C$

$$\therefore 2C + C = \pi \quad \therefore C = \frac{\pi}{3}]$$

5. If $A + B + C = 2\pi$, prove that

$$\cos^2B + \cos^2C - \sin^2A - 2\cos A \cos B \cos C = 0$$

6. (i) If $A + B + C = 0$, prove that

$$\cos^2A + \cos^2B + \cos^2C = 1 + 2\cos A \cos B \cos C$$

(ii) Prove that $\cos^2(\beta - \gamma) + \cos^2(\gamma - \alpha) + \cos^2(\alpha - \beta) = 1 + 2\cos(\beta - \gamma) \cos(\gamma - \alpha) \cos(\alpha - \beta)$

(iii) If $A + B + C = \pi$, prove that

$$\sin A \cos B \cos C + \sin B \cos C \cos A + \sin C \cos A \cos B = \sin A \sin B \sin C$$

[Hint : divide both sides by $\sin A \sin B \sin C$]

Type II.

7. If $A + B + C = \frac{\pi}{2}$, show that

$$(i) \cot A + \cot B + \cot C = \cot A \cot B \cot C$$

$$(ii) \tan A \tan B + \tan B \tan C + \tan C \tan A = 1$$

8. If $A + B + C = \pi$, prove that

$$(i) \tan 3A + \tan 3B + \tan 3C = \tan 3A \tan 3B \tan 3C \quad [\text{S. C. R. A. 75}]$$

$$(ii) \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$$

$$(iii) \frac{\cot A + \cot B}{\tan A + \tan B} + \frac{\cot B + \cot C}{\tan B + \tan C} + \frac{\cot C + \cot A}{\tan C + \tan A} = 1$$

$$[\text{Hint : Put } \tan A = \frac{1}{\cot A}, \tan B = \frac{1}{\cot B}, \tan C = \frac{1}{\cot C}]$$

Conditional Trigonometrical Identities

115

9. Prove that $\tan(\alpha - \beta) + \tan(\beta - \gamma) + \tan(\gamma - \alpha)$
 $= \tan(\alpha - \beta) \tan(\beta - \gamma) \tan(\gamma - \alpha)$ [C. E. E. (Bihar) '85]

[Hint : Let $A = \alpha - \beta$, $B = \beta - \gamma$, $C = \gamma - \alpha$, then $A + B + C = 0$]

10. If $x + y + z = 0$ show that $\cot(x + y - z) \cot(z + x - y)$
 $+ \cot(x + y - z) \cot(y + z - x) + \cot(y + z - x) \cot(y + x - y) = 1$

[Hint: Let $A = x + y - z$, $B = z + x - y$, $C = y + z - x$, then

$$A + B + C = x + y + z = 0]$$

11. If $A + B + C = n\pi$ (n being zero or an integer)

Show that $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

[Hint : $A + B = n\pi - C$ and $\tan(n\pi - C) = -\tan C$

$$\therefore \tan(\pi + \theta) = \tan \theta \text{ and } \tan(\pi - \theta) = -\tan \theta]$$

Type III.

12. If $A + B + C = \pi$, prove that

(i) $\cos 2A + \cos 2B - \cos 2C = 1 - 4\sin A \sin B \cos C$

(ii) $\sin 2A + \sin 2B - \sin 2C = 4\cos A \cos B \sin C$

(iii) $\sin A + \sin B + \sin C = 4\cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$

(iv) $\cos A + \cos B - \cos C = 4\cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2} - 1$ [Roorkee 66]

(v) $\sin(B + C - A) + \sin(C + A - B) + \sin(A + B - C)$
 $= 4\sin A \sin B \sin C$ [S. C. R. A. 74]

[Hint : $(B + C) - A = (\pi - A) - A = \pi - 2A$

$$\therefore \sin(B + C - A) = \sin(\pi - 2A) = \sin 2A \text{ etc.}$$

Now L.H.S. = $\sin 2A + \sin 2B + \sin 2C$ then proceed as in W. O.

Ex. (10)]

(vi) $\frac{\cos A}{\sin B \sin C} + \frac{\cos B}{\sin C \sin A} + \frac{\cos C}{\sin A \sin B} = 2$

[Hint : L.H.S. = $\frac{\cos A \sin A + \cos B \sin B + \cos C \sin C}{\sin A \sin B \sin C}$

$$= \frac{\sin 2A + \sin 2B + \sin 2C}{2\sin A \sin B \sin C}$$

Now show that $\sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C$

(vii) $\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = 8\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

13. If $x + y + z = \frac{\pi}{2}$, prove that

$$\cos(x - y - z) + \cos(y - z - x) + \cos(z - x - y) - 4\cos x \cos y \cos z = 0$$

$$\left[\text{Hint : } x - y - z = x - \left(\frac{\pi}{2} - x \right) = 2x - \frac{\pi}{2} = - \left(\frac{\pi}{2} - 2x \right) \right]$$

14. Show that $\sin(x - y) + \sin(y - z) + \sin(z - x)$

$$+ 4\sin \frac{x-y}{2} \sin \frac{y-z}{2} \sin \frac{z-x}{2} = 0$$

15. If $A + B + C = 180^\circ$, prove that

$$\sin(B + 2C) + \sin(C + 2A) + \sin(A + 2B)$$

$$= 4\sin \frac{B-C}{2} \sin \frac{C-A}{2} \sin \frac{A-B}{2} \quad [\text{Roorkee 67}]$$

16. If $A + B + C = \pi$, prove that

$$\sin \frac{B+C}{2} + \sin \frac{C+A}{2} + \sin \frac{A+B}{2}$$

$$= 4\cos \frac{\pi-A}{4} \cos \frac{\pi-B}{4} \cos \frac{\pi-C}{4}$$

$$\left[\text{Hint : L.H.S.} = \sin \left(\frac{\pi}{2} - \frac{A}{2} \right) + \sin \left(\frac{\pi}{2} - \frac{B}{2} \right) + \sin \left(\frac{\pi}{2} - \frac{C}{2} \right) \right]$$

$$= \cos \frac{A}{2} + \cos \frac{B}{2} + \cos \frac{C}{2}$$

Type IV.

17. If $x + y + z = xyz$, prove that

$$(i) \frac{2x}{1-x^2} + \frac{2y}{1-y^2} + \frac{2z}{1-z^2} = \frac{2x}{1-x^2} \frac{2y}{1-y^2} \frac{2z}{1-z^2}$$

$$(ii) x(1-y^2)(1-z^2) + y(1-z^2)(1-x^2) + z(1-x^2)(1-y^2) = 4xyz.$$

Type V.

18. If $A + B + C + D = 2\pi$, show that

$$\cos A - \cos B + \cos C - \cos D = 4\sin \frac{A+B}{2} \sin \frac{A+D}{2} \cos \frac{A+C}{2}$$

[Roorkee 70]

19. If A, B, C, D be the angles of a cyclic quadrilateral, show that $\cos A + \cos B + \cos C + \cos D = 0$ [C. E. E. (West Bengal) 83]

20. If $A + B + C = \pi$, prove that

$$\cos A \sin B \sin C + \cos B \sin C \sin A + \cos C \sin A \sin B$$

$$= 1 + \cos A \cos B \cos C$$

Conditional Trigonometrical Identities



21. If $A + B + C = \pi$, prove that $\cot^2 A + \cot^2 B + \cot^2 C \geq 1$

[Hint : $(\cot A - \cot B)^2 + (\cot B - \cot C)^2 + (\cot C - \cot A)^2 \geq 0$

and $A + B + C = \pi$, $\therefore \cot A \cot B + \cot B \cot C + \cot C \cot A = 1$]

22. If $A + B + C = \pi$, prove that

$$\begin{aligned} \cos \frac{A}{2} \cos \frac{B-C}{2} + \cos \frac{B}{2} \cos \frac{C-A}{2} + \cos \frac{C}{2} \cos \frac{A-B}{2} \\ = \sin A + \sin B + \sin C \quad \text{[S. C. R. A. 71]} \end{aligned}$$

$$\begin{aligned} \text{[Hint : } \cos \frac{A}{2} \cos \frac{B-C}{2} &= \cos \left(\frac{\pi}{2} - \frac{B+C}{2} \right) \cos \frac{B-C}{2} \\ &= \sin \frac{B+C}{2} \cos \frac{B-C}{2} = \frac{1}{2} \left(2 \sin \frac{B+C}{2} \cos \frac{B-C}{2} \right) \\ &= \frac{1}{2} (\sin B + \sin C) \text{ etc.}] \end{aligned}$$

23. In a ΔABC , prove that

$$\sin 3A \sin(B-C) + \sin 3B \sin(C-A) + \sin 3C \sin(A-B) = 0$$

[S. C. R. A. 74]

24. In a ΔABC , prove that

$$\begin{aligned} \sin^3 A \cos(B-C) + \sin^3 B \cos(C-A) + \sin^3 C \cos(A-B) \\ = 3 \sin A \sin B \sin C \end{aligned}$$

$$\begin{aligned} \text{[Hint : } \sin^3 A \cos(B-C) &= \sin^2 A \sin A \cos(B-C) \\ &= \frac{1}{2} \sin^2 A [2 \sin(B+C) \cos(B-C)] \quad [\because A = \pi - (B+C)] \\ &= \frac{1}{2} \sin^2 A [\sin 2B + \sin 2C] \\ &= \frac{1}{2} \sin^2 A [2 \sin B \cos B + 2 \sin C \cos C] \end{aligned}$$

